

Regression Metrics Master Guide (Telugu Transliteration)

Ee document lo, "RegressionMetrics.pdf" lo unna concepts ni base chesukuni, Regression models ni evaluate cheyadaniki use ayye mukhya metrics (MAE, MSE, RMSE, R^2 mariyu Adjusted R^2) gurinchi pin-to-pin Telugu transliteration lo (English letters tho) chala detailed ga explain chestanu. Meeru nidra lo lechina error lekunda code rayagalige antha strong coding practice mariyu visualizations ni ikkada include chesanu.

1. Regression Metrics Parichayam (Evaluation Metrics Ante Enti?)

Oka Machine Learning model (e.g., Linear Regression) ni train chesina tarwata, adi entha бага pani chestundi ani telusukovalante manam **Evaluation Metrics** ni use chestaru. Ivi oka model ki iche **report card** lantivi.

- Question:** Mee model predict chesina values actual values ki entha daggaraga unnayi?
- Answer:** Regression metrics manaku exact ga aa error ni mariyu fit ni number rupam lo cheptayi.

2. Goodness-of-Fit Metrics: R^2 (R-Squared) mariyu Adjusted R^2

A. R^2 (R-Squared / Coefficient of Determination)

R^2 ante enti?

R^2 ante target variable lo unna total variation lo, mee independent variables (features) dwara **entha percent variation explain avtundi** ani cheppedi. Idi model data ni entha бага fit chesindo chupistundi.

Formula:

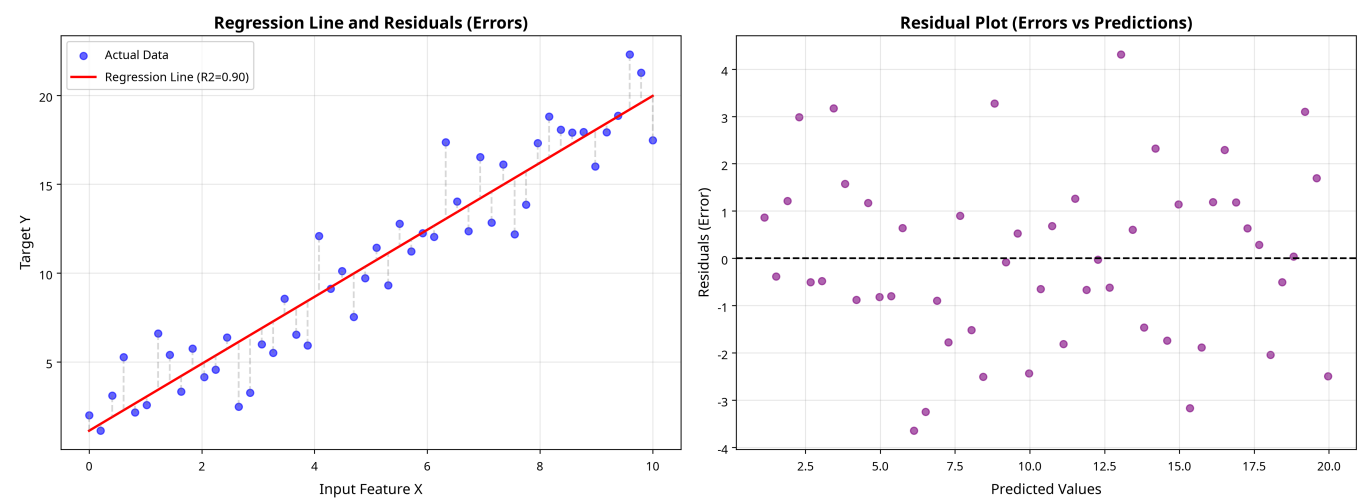
$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

- y_i = Actual value (nijamaina value)
- $\hat{y}_i$ = Predicted value (model predict chesina value)
- $\bar{y}$ = Mean of actual values (actual values average)

- **SSR (Sum of Squared Residuals)** = Model chestunna errors yokka squares sum.
- **SST (Total Sum of Squares)** = Actual values ki mean ki madhya unna total variation.

Visualization: Regression Line and Residuals

Kinda unna plot lo actual data points, regression line, mariyu errors (residuals) ela untayo chudandi:



R² Interpretation (R² values ki ardam ento?)

R ² Value	Ardam (Meaning)
1.0	Perfect prediction! Errors assalu levu.
0.95	Excellent model fit.
0.80	Good model fit (80% variation explained).
0.50	Moderate model.
0.0	Model em nerchukoledu, adi kevalam mean value ni predict chestundi.
Negative (< 0)	Model mean ni predict cheyadam kante worst ga pani chestundi!

R² yokka Pedda Problem (The Limitation of R²)

R² ki oka bad habit undi: meeru model ki enni **useless features** (e.g., mobile number, pin code, random noise) add chesina, **R² eppudu perugutundi leda alage untundi, kani eppatiki taggadu!** Idi model overfitting ni hide chestundi. Ee problem ni solve cheyadaniki **Adjusted R²** vadatam.

B. Adjusted R^2 (Adjusted R-Squared)

Adjusted R^2 ante enti?

Adjusted R^2 ante R^2 yokka improved version. Idi model ki kotha feature add chesinappudu, adi nijam ga use avtunda leda ani check chesi penalty istundi. Useless features add chesthe Adjusted R^2 taggutundi.

Formula:

$$\text{Adjusted } R^2 = 1 - \left[(1 - R^2) \times \frac{n - 1}{n - p - 1} \right]$$

- n = Number of observations (total rows)
- p = Number of independent variables (features sankhya)

Rule of Thumb:

- Useful feature add chesthe $\rightarrow$ Adjusted R^2 perugutundi.
- Useless feature add chesthe $\rightarrow$ Adjusted R^2 taggutundi.
- Eppudu **Adjusted $R^2 \leq R^2$** untundi.

3. Error-Based Metrics (MAE, MSE, RMSE)

Model chestunna errors ni direct ga kolavadaniki ee 3 metrics vadatam:

1. **MAE (Mean Absolute Error):** Errors yokka absolute values average. Outliers ki less sensitive. $\text{MAE} = \frac{1}{n} \sum |y_i - \hat{y}_i|$
 2. **MSE (Mean Squared Error):** Errors yokka squared values average. Outliers ki chala sensitive (heavy penalty). $\text{MSE} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$
 3. **RMSE (Root Mean Squared Error):** MSE ki square root. Idi actual target variable unna units lo ne untundi (e.g., Rupees, Lakhs), anduke ardham chesukovadaniki chala easy. $\text{RMSE} = \sqrt{\text{MSE}}$
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4. MASTER CODING SECTION: Step-by-Step Python Implementation

Meeru error lekunda code rayadaniki, ikkada scikit-learn use chesi regression metrics calculate chese complete Python code ichchanu.

```

# =====
# STEP 1: Import Required Libraries
# =====
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# =====
# STEP 2: Create / Load Dataset
# =====
np.random.seed(42)
# House size (sq ft) and number of rooms as features (p = 2)
size = np.random.randint(500, 3000, 100)
rooms = np.random.randint(1, 5, 100)

# Price = 20 + 0.1 * size + 5 * rooms + random noise
price = 20 + 0.1 * size + 5 * rooms + np.random.normal(0, 10, 100)

df = pd.DataFrame({'Size': size, 'Rooms': rooms, 'Price': price})

# =====
# STEP 3: Define X (Features) and Y (Target)
# =====
X = df[['Size', 'Rooms']] # p = 2 features
Y = df['Price']

# =====
# STEP 4: Train-Test Split
# =====
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)

# =====
# STEP 5: Train Model
# =====
model = LinearRegression()
model.fit(X_train, Y_train)

# Predictions
Y_pred = model.predict(X_test)

# =====

```

```
# STEP 6: Calculate Regression Metrics
# =====
mae = mean_absolute_error(Y_test, Y_pred)
mse = mean_squared_error(Y_test, Y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(Y_test, Y_pred)

# Calculate Adjusted R2
n = X_test.shape[0] # Number of observations
p = X_test.shape[1] # Number of features
adjusted_r2 = 1 - ((1 - r2) * (n - 1) / (n - p - 1))

print("--- REGRESSION EVALUATION METRICS ---")
print(f"Mean Absolute Error (MAE): {mae:.2f}")
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.2f}")
print(f"R2 Score: {r2:.4f}")
print(f"Adjusted R2 Score: {adjusted_r2:.4f}")
```

5. Real-Life & Data Science Applications

- **House Price Prediction:** R^2 mariyu RMSE use chesi model actual prices ni entha accurate ga predict chestundo check chestaru.
- **Stock Market Forecasting:** Financial prices lo MSE mariyu MAE use chesi error magnitude ni measure chestaru.
- **Feature Selection:** Multiple regression lo useless features ni remove cheyadaniki Adjusted R^2 ni primary tool ga vadataru.

AI/ML/DL Model Recommendations

Ee metrics anni **Linear Regression, Ridge/Lasso Regression, Support Vector Regressor (SVR), Random Forest Regressor, XGBoost**, mariyu **Neural Networks (Regression tasks)** lo model performance ni evaluate cheyadaniki standard ga vadataru.

References:

1. [RegressionMetrics.pdf](#) - Original PDF document for R^2 and Adjusted R^2 evaluation metrics.